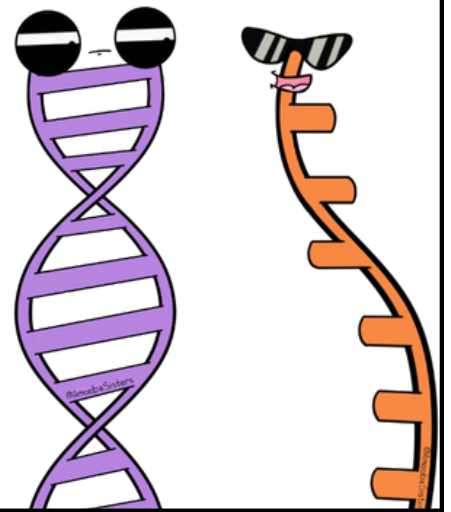


Amoeba Sisters Video Recap: DNA vs. RNA & Protein Synthesis

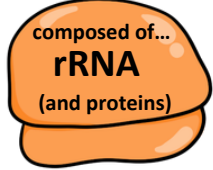
Whose Show Is This?

DNA shouldn't get all the credit! For this portion, check out the Amoeba Sisters DNA vs. RNA video. Then, write "D" if for DNA, "R" if for RNA, or "BOTH" if it pertains to both DNA and RNA.

1. _____ I am a **nucleic acid**.
2. _____ I am usually **single-stranded**.
3. _____ I am generally found both inside and outside of the **nucleus** [in eukaryotic cells].
4. _____ I am arranged as a **double helix**, and my shape is often described as a "**twisted ladder**."
5. _____ I include bases **guanine**, **cytosine**, and **adenine**.
6. _____ Each of my nucleotides includes a **phosphate**, **sugar**, and **base**.
7. _____ I include the base **uracil**.
8. _____ I include the base **thymine**.
9. _____ I generally remain in the **nucleus** [in eukaryotic cells].
10. _____ I have the sugar **deoxyribose**.
11. _____ I am made up of **nucleotides**.
12. _____ I have the sugar **ribose**.



For the following discussed RNA types, complete the missing information in the boxes below. Some boxes have been filled in for you.

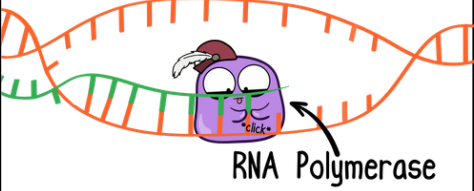
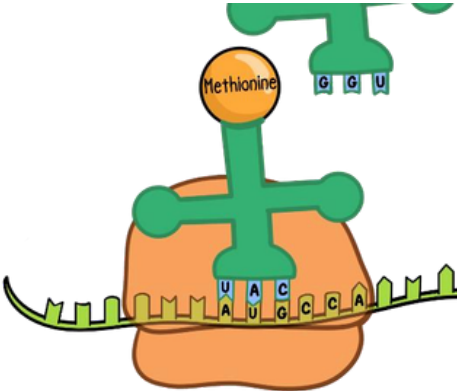
Type: <i>mRNA</i>	13. Type: _____	14. Type: _____
Stands for:	Stands for:	Stands for:
15. _____	16. _____	<i>Transfer RNA</i>
Sketch to Help you Remember:	Sketch to Help you Remember:	Sketch to Help you Remember:
17. _____		18. _____
General Function:	General Function:	General Function:
19. _____	20. _____	<i>Transfers amino acids [to area of protein synthesis].</i>



Amoeba Sisters Video Recap: DNA vs. RNA & Protein Synthesis

Protein Synthesis Summary

Complete the missing information in the summary chart after watching the Amoeba Sisters Protein Synthesis video.

Process Name	Location (in eukaryotic cell)	Brief and General Description	End Result	DNA <i>directly</i> involved? (yes or no?)	List RNA type(s) involved (mRNA, rRNA, and/or tRNA?)
Transcription 	21.	22.	23.	24.	mRNA Only
Translation 	25.	26.	27.	No	28.

29. Consider the illustration placed in the *transcription* box above. Identify and label on the illustration *if* any of the following are present: **DNA, mRNA, rRNA, tRNA, and/or amino acid.**

30. Consider the illustration placed in the *translation* box above. Identify and label on the illustration *if* any of the following are present: **DNA, mRNA, rRNA, tRNA, and/or amino acid.**

